

SIMATS ENGINEERING

CO4 – AT1 - Article Review- Low

**COURSE NAME: Advance Wireless Communication Systems / 5G / 6G
Communication Systems**

COURSE CODE: ECA56

COURSE OUTCOME COVERED

CO: 5 - Develop 5G-enabled edge computing and IoT solutions for smart city, healthcare, industrial automation, and intelligent communication applications. (BL6)

Assessment Tool Weightage: 25%

Short description about assessment: Students review recent technical articles to assess critical thinking and awareness of trends.

Dr. KALAIARASI

Total Marks: Duration: 90 Minutes

QUESTIONS

Edge AI for Intelligent 5G Network Optimization (BL5)

“Service-aware real-time slicing for virtualized beyond 5G networks” — 2024

- **Authors:** Theodoros Tsourdinis, Ilias Chatzistefanidis, Nikos Makris, Thanasis Korakis, Navid Nikaein, Serge Fdida
- **Journal:** *Computer Networks*, Volume 247
- **Year:** 2024
- **DOI:** 10.1016/j.comnet.2024.110445
- **Publication:** June 2024
- **Publisher:** Elsevier

https://www.sciencedirect.com/science/article/pii/S1389128624002779?utm_source=chatgpt.com

Abstract

This paper presents a **service-aware 5G network architecture based on Edge Intelligence and Machine Learning (ML)** to improve real-time network management and resource allocation. Traditional 5G networks generally respond to changes in network conditions after they occur. The proposed approach instead predicts the future requirements of network traffic and proactively adjusts the network resources. ML models are used to classify cellular traffic flows and predict their effect on network load. Based on these predictions, the system automatically reallocates **Radio Access Network (RAN) slices** according to the requirements of different applications. The framework also incorporates distributed and online MLOps techniques, allowing the ML models to continuously adapt to changing network conditions. Experiments were performed in a realistic 5G environment with measurements at a granularity of **10 ms**. The proposed AI/ML-based approach achieved up to **4× lower latency (jitter)** and nearly **4× higher throughput** compared with the non-AI/ML baseline under the studied conditions.

Technology Implemented

The paper implements a **cloud-native, Edge-AI-driven 5G network slicing framework** that combines Machine Learning, network analytics, virtualization, and automated resource allocation. The main technologies are:

- **OpenAirInterface (OAI)**

The researchers use **OpenAirInterface (OAI)** to implement the cellular network components. OAI provides the RAN and core-network functionality and allows the researchers to experiment with network slicing and programmable network functions. The network functions are deployed as software components rather than relying entirely on dedicated hardware.

- **Kubernetes-based Cloud-Native Architecture**

The complete 5G network is deployed using **Kubernetes**. Network functions are organized as containers/microservices, allowing them to be monitored, scaled, and managed dynamically. This is important for Edge AI because computing resources can be distributed across different nodes according to the workload.

- **Network Data Analytics Function (NWDAF)**

A customized analytics function called **Core RAN Analytics Function (CRAF)** is developed because an open-source NWDAF implementation was not available for their setup. It collects network information from the core and RAN and stores metrics such as:

- **Throughput**
- **Jitter**
- **Channel Quality Indicator (CQI)**
- Traffic information
- User/application information

These measurements provide the data required by the AI models to understand network conditions.

- **Machine Learning for Traffic Classification**

Machine Learning is used to identify and classify different cellular traffic flows according to the applications generating them. Instead of simply reacting to existing network congestion, the system uses the traffic patterns to **predict future network requirements**. Six different neural-network-based approaches were evaluated to identify an appropriate model.

- **Edge Intelligence**

The AI/ML processing is distributed toward the **network edge**, close to the RAN. This reduces dependence on a centralized cloud and enables decisions to be made with very low delay. The system was designed to support network monitoring and decision-making with a granularity of approximately **10 ms**.

- **Dynamic RAN Network Slicing**

The most important technology in the paper is **dynamic RAN slicing**. Network resources are divided into virtual slices, with each slice serving different application requirements. The ML system predicts the requirements of traffic flows and the framework uses programmable APIs to dynamically adjust slice allocation. This helps prevent both **over-provisioning** and **under-provisioning** of resources.

- **MLOps and Distributed Training**

The researchers implement an **MLOps architecture** using distributed and online training. **Kubeflow pipelines** are used to orchestrate ML tasks running as containerized components. This allows the model to continuously receive new network data, update itself, and adapt to changing traffic patterns instead of relying only on a fixed training dataset.

- **Real-Time Network Monitoring**

The framework continuously monitors network performance and user traffic. Realistic mobility scenarios are created using programmable attenuators, while real devices generate application traffic. This allows the AI system to observe changes in network conditions and make corresponding resource-allocation decisions.

Technology Flow

User/Application Traffic → Network Monitoring → CRAF/NWDAF → Data Processing → ML Traffic Classification & Prediction → Edge AI Decision → Dynamic RAN Slice Allocation → Improved QoS

Conclusion

The paper demonstrates that combining **Edge Intelligence, Machine Learning, network slicing, and cloud-native 5G technologies** can significantly improve network performance. Instead of allocating fixed resources, the proposed system predicts application traffic requirements and dynamically adjusts RAN

slices. This reduces unnecessary resource allocation while ensuring that applications receive the resources they require. The experimental results demonstrate improvements in both **latency and throughput**, while also reducing over-provisioning and under-provisioning of network resources and improving energy efficiency. The work therefore provides a strong foundation for **AI-driven autonomous 5G and beyond-5G network optimization**. It is directly relevant to the project “**Edge AI for Intelligent 5G Network Optimization**” because it demonstrates how AI at the network edge can monitor traffic, predict network demand, and automatically optimize 5G resources in near real time.

Reference:

Tsourdinis, T., Chatzistefanidis, I., Makris, N., Korakis, T., Nikaein, N., & Fdida, S. (2024). *Service-aware real-time slicing for virtualized beyond 5G networks*. **Computer Networks**, **247**, 110445.

Code of Conduct:

I ANNIE RAKSHITHA J (192512442) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Annie Rakshitha

RUBRICS FOR CALCULATION:

Article Review					
Criteria	Weight ag e	Level 4 (Exemplar y)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understan din g of Article	20%	Demonstra tes clear and in-depth understand ing of the article’s purpose, concepts, and arguments	Shows good understand ing with minor gaps	Partial understandi ng; misses key ideas	Lacks understand ing of the article
Summary	20%	Provides concise and accurate summary highlightin g all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with	Provides reasonable analysis with some	Superficial analysis; limited evaluation	No meaningful analysis

		strong reasoning and evaluation	justification		
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand